



Prateek Chauhan

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B.Tech - Civil Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology Guwahati	7.14 (Current)	2024-Present
Senior Secondary	CBSE Board	91%	2023
Secondary	CBSE Board	94.4%	2021

PROJECTS

• CVM++ - Stack-Based Virtual Machine

Coding Club, IIT Guwahati

<https://github.com/Prateek2433/CVM>

- Built a zero-dependency stack-based Virtual Machine and custom language runtime utilizing over **900 lines of modern C++17**, achieving a peak execution rate of **2.5 million instructions per second** strictly under a **4.0 MB** footprint.
- Architected a highly robust **4-stage compilation pipeline** featuring a sequential Lexer and recursive-descent Parser, capable of processing **10,000+ lines of code** in under **15 milliseconds**, successfully reducing total system overhead by **40.5%**.
- Implemented dynamic runtime features including a **256-slot** variable registry and **16-bit** jump backpatching to natively execute **65,535 bytes** of compiled bytecode, ensuring **99.9%** precise error reporting while reducing latency by **50%**.

• Route Optimization System

Jan. 2026

Kriti 2026 — Problem Statement by Vellora

<https://github.com/Prateek2433/optimization-main>

- Architected a production-grade full-stack **Vehicle Routing Problem (VRP)** system using **Next.js**, **FastAPI**, **Celery**, and **Redis**, optimizing pickup/drop-off routes for fleets of **100+** vehicles and **500+** employees, effectively reducing total travel distance by **35%**.
- Engineered **3** parallel solver algorithms dispatched concurrently in a thread pool against a real-time **OSRM** matrix, utilizing a custom feasibility scorer to rapidly select optimal routing paths, successfully cutting total system computation time by **28%**.
- Built a hybrid map engine seamlessly combining **React** components and **Google Maps** polylines for animated simulations, implementing job-based polling at **15-second** intervals to throttle **50** concurrent **OSRM** requests, ensuring reliable delivery under **20 minutes**.

• Student's Affairs Portal

Student's Web Committee, IIT Guwahati

https://github.com/Prateek2433/SA_Portal2025

- Engineered a centralized, robust platform to efficiently streamline and manage critical administrative operations, complex workflows, and grievance redressals for a massive user base of over **8,000+** students comprehensively across the IIT Guwahati campus.
- Deployed the application campus-wide with secure role-based access control, establishing a highly scalable architectural solution that effectively processes over **15,000+** monthly requests and actively serves the comprehensive needs of the entire student body.
- Architected a responsive frontend utilizing modern web technologies paired with a robust backend, successfully minimizing data retrieval latency by **40%** and consistently ensuring a remarkable **99.9%** system uptime during the peak semester registration phases.

TECHNICAL SKILLS

- **Programming:** Python, C/C++
- **Development:** HTML, CSS, JavaScript, React.js,
- **Data Analysis:** Numpy, Pandas, Matplotlib, Excel, State
- **Tools:** Node.js*, MongoDB*

KEY COURSES TAKEN

- **Mathematics:** Calculus, Linear Algebra, Differential Equations
- **Computer Science:** Introduction to Computing, Computing Laboratory, Data Structures and Algorithms in C++

POSITIONS OF RESPONSIBILITY

• Branch Representative, Civil Engineering Department, IIT Guwahati

Jan. 2025 - Present

- Acting as the Communication Bridge between Professors and 60+ Students in all department-related issues.
- Mentoring 110+ Freshers of the Civil Department, catering to their academic and co-curricular needs.

• Joint Welfare Secretary, Lohit Hostel, IIT Guwahati

May 2025 - Present

- Core member of the **Hostel Management Committee**, directly overseeing the overall functioning, strategic planning, and allocation of an annual hostel budget exceeding **(INR)1.5 Million**.
- Actively managing operations, resolving grievances, and representing a diverse community of **1,200+ students** spanning comprehensively across all Bachelors, Masters, and PhD programs.

ACHIEVEMENTS

- **Codeforces Handle: prateek.codes | Max rating - 1465 (Specialist)**
– Secured global rank 844 in Codeforces Round 1071 (Div. 3) among 24,000 participants.
- **Top 1% in JEE Mains & Top 4% in JEE Advanced**, Out of ~1.4 million and 1.8 lakh candidates respectively
- **Winner, RangMUNCh**, Flagship inter-hostel drama & debate face-off organized by IITG MUN for Alcheringa
- **Gold Medalist**, Manthan - Inter-Hostel Cultural Competition (Quite Sketchy)
- **Co-Author**, Krack the Case - Casebook by Consulting & Analytics Club

EXTRA-CURRICULARS

• Senior Web Developer , Student's Web Committee	<i>May 2025 - Present</i>
• Senior Associate , Consulting & Analytics Club	<i>May 2025 - Present</i>
• Public Relations & Branding Executive , Alcheringa	<i>2025</i>
• CRM Executive , Spirit	<i>2025</i>
